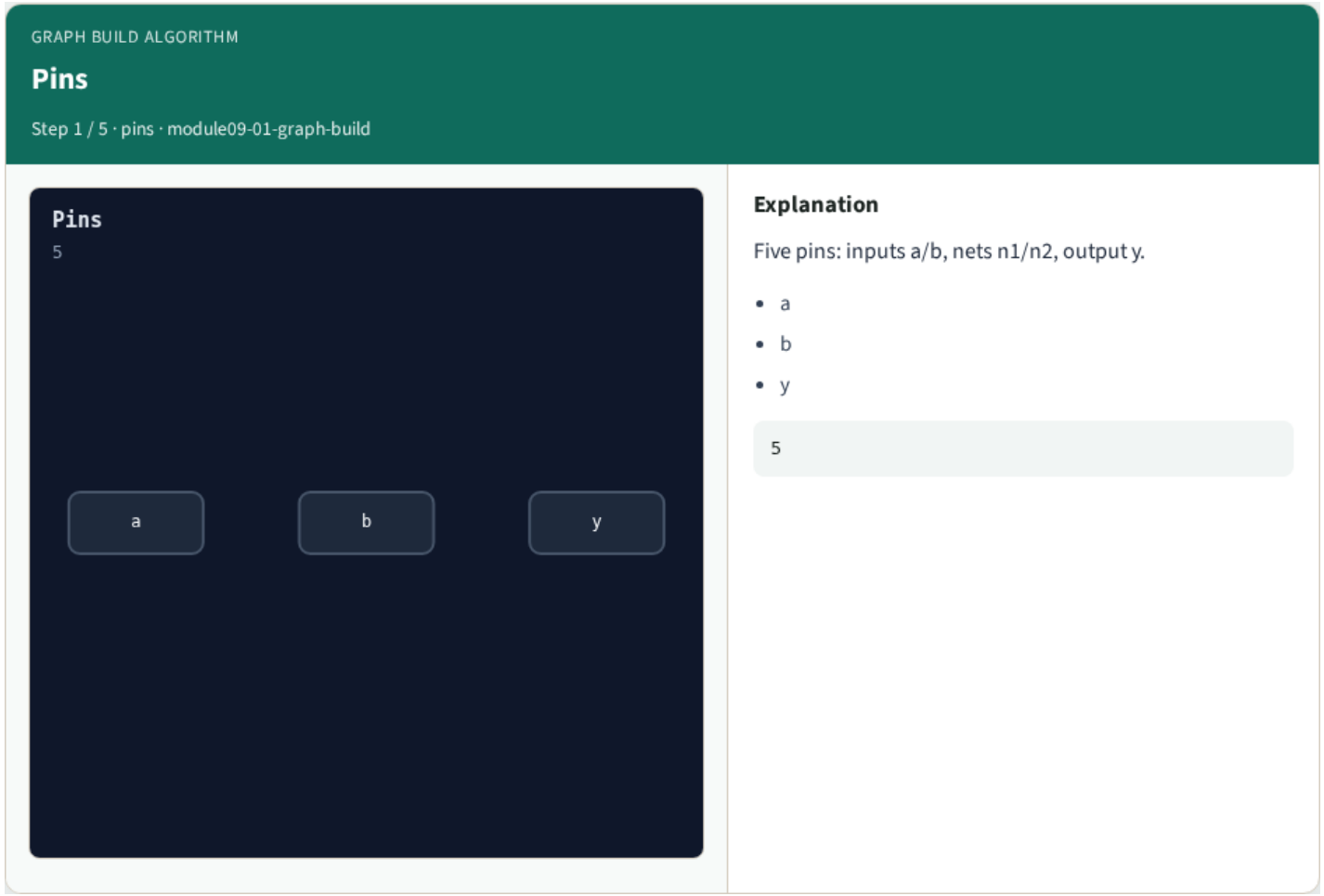
Why algorithms

Build the timing graph from pins and delay arcs

The idea

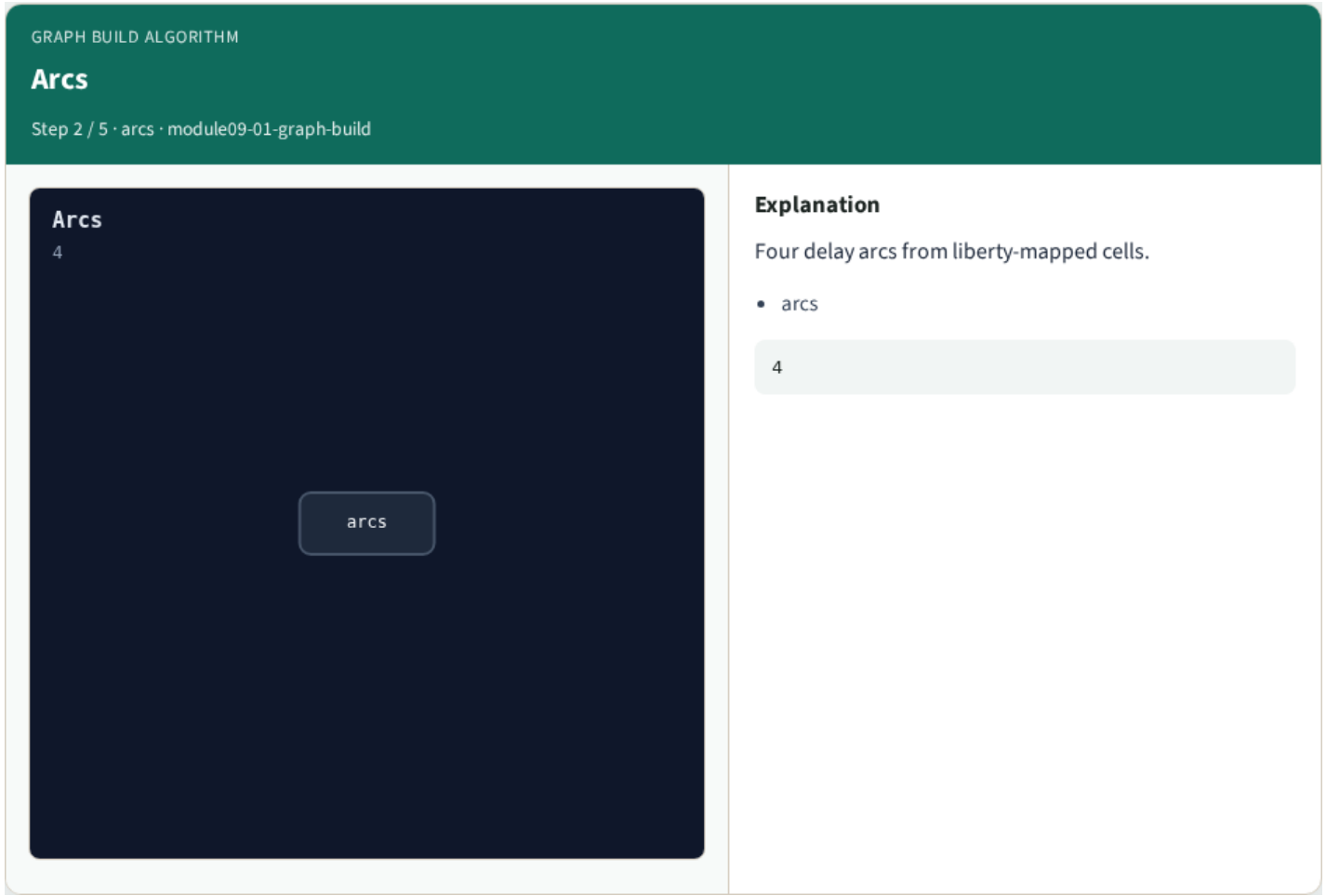
- GOLDEN: 5 pins, 4 arcs; cells u1/u2/u3

Pins


GRAPH BUILD ALGORITHM
Pins
Step 1 / 5 · pins · module09-01-graph-build
Pins
5
a
b
y
Explanation
Five pins: inputs a/b, nets n1/n2, output y.
a
b
y
5

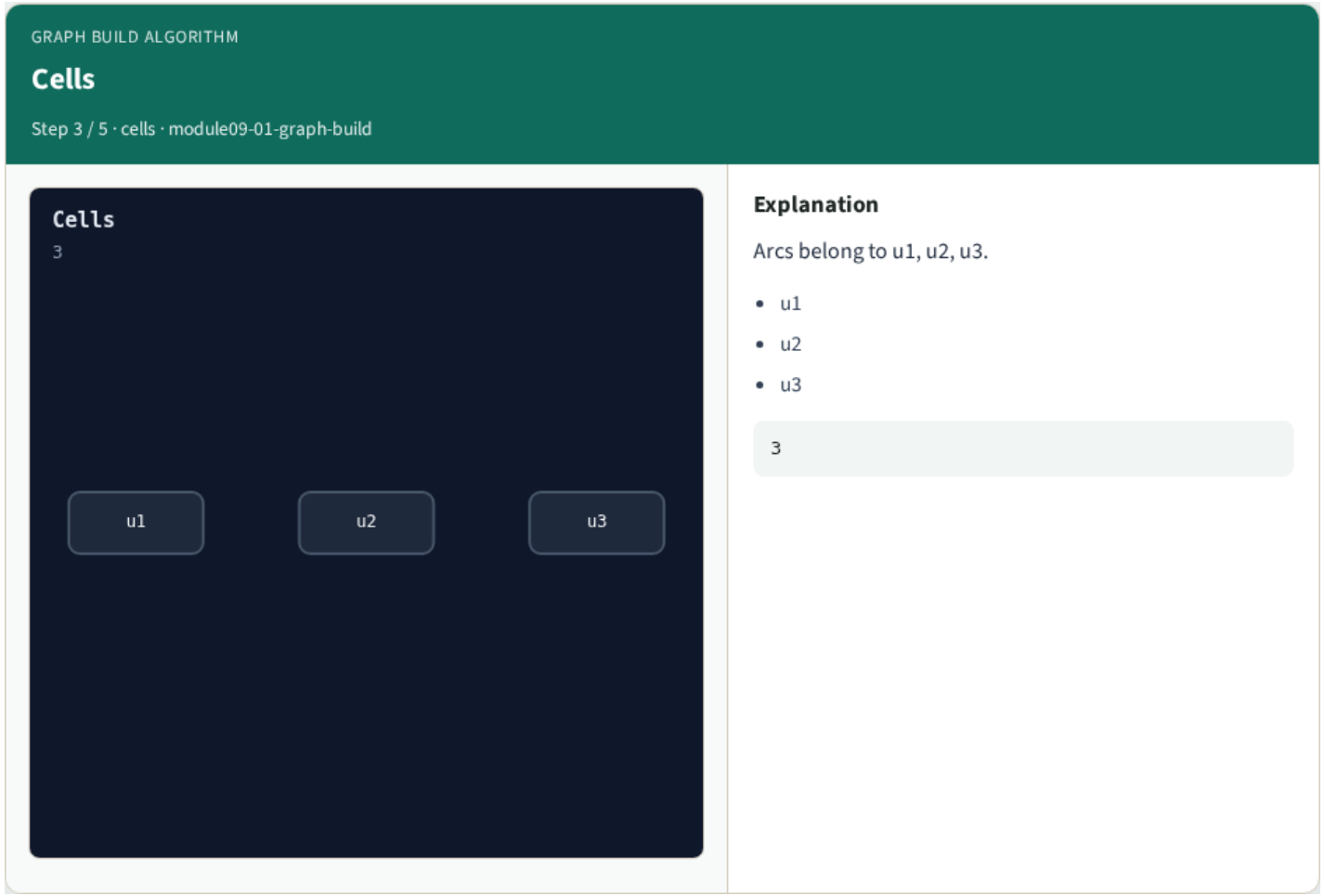
Pins

Arcs


GRAPH BUILD ALGORITHM
Arcs
Step 2 / 5 · arcs · module09-01-graph-build
Arcs
4
arcs
Explanation
Four delay arcs from liberty-mapped cells.
arcs
4

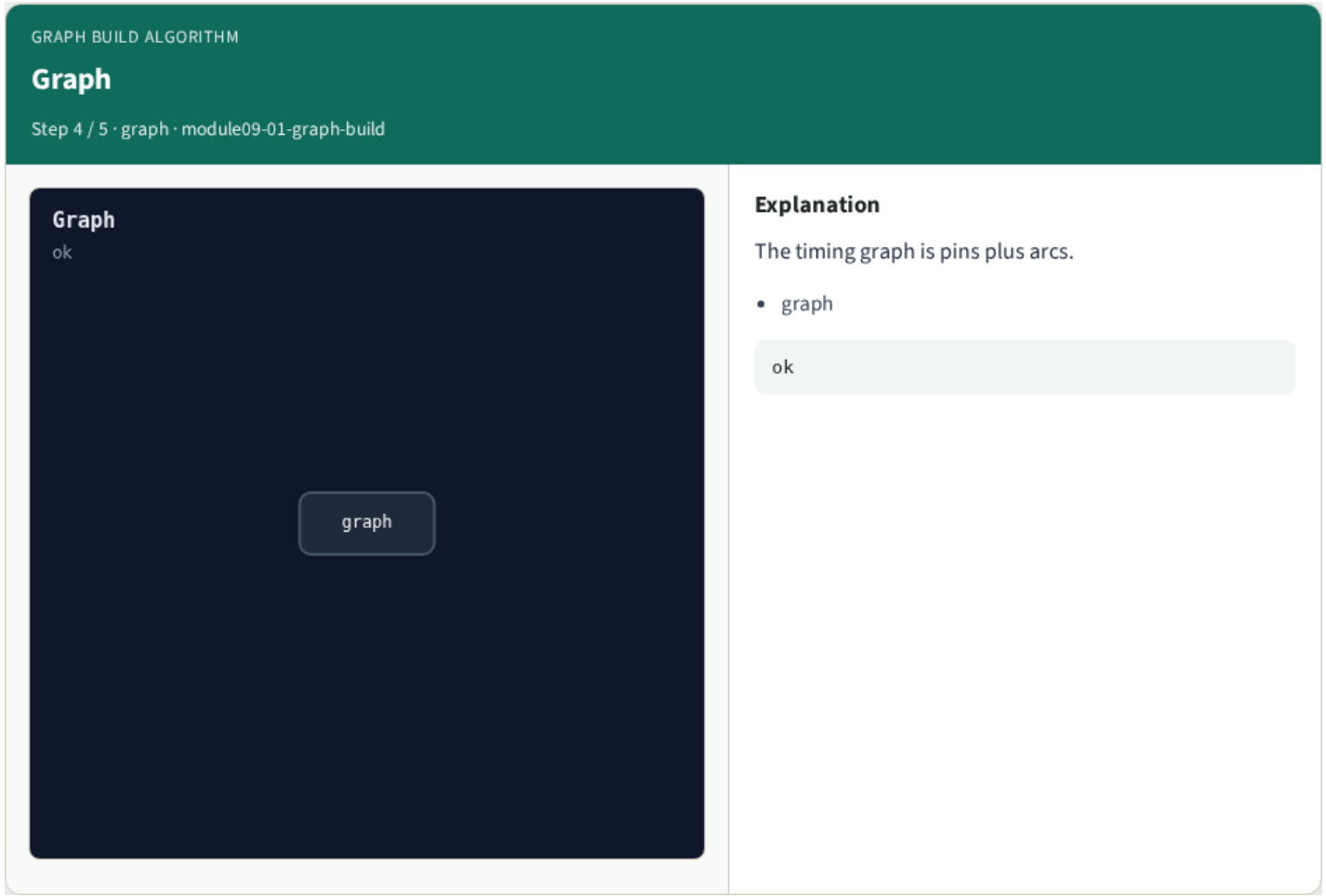
Arcs

Cells


GRAPH BUILD ALGORITHM
Cells
Step 3 / 5 · cells · module09-01-graph-build
Cells
3
u1
u2
u3
Explanation
Arcs belong to u1, u2, u3.
u1
u2
u3
3

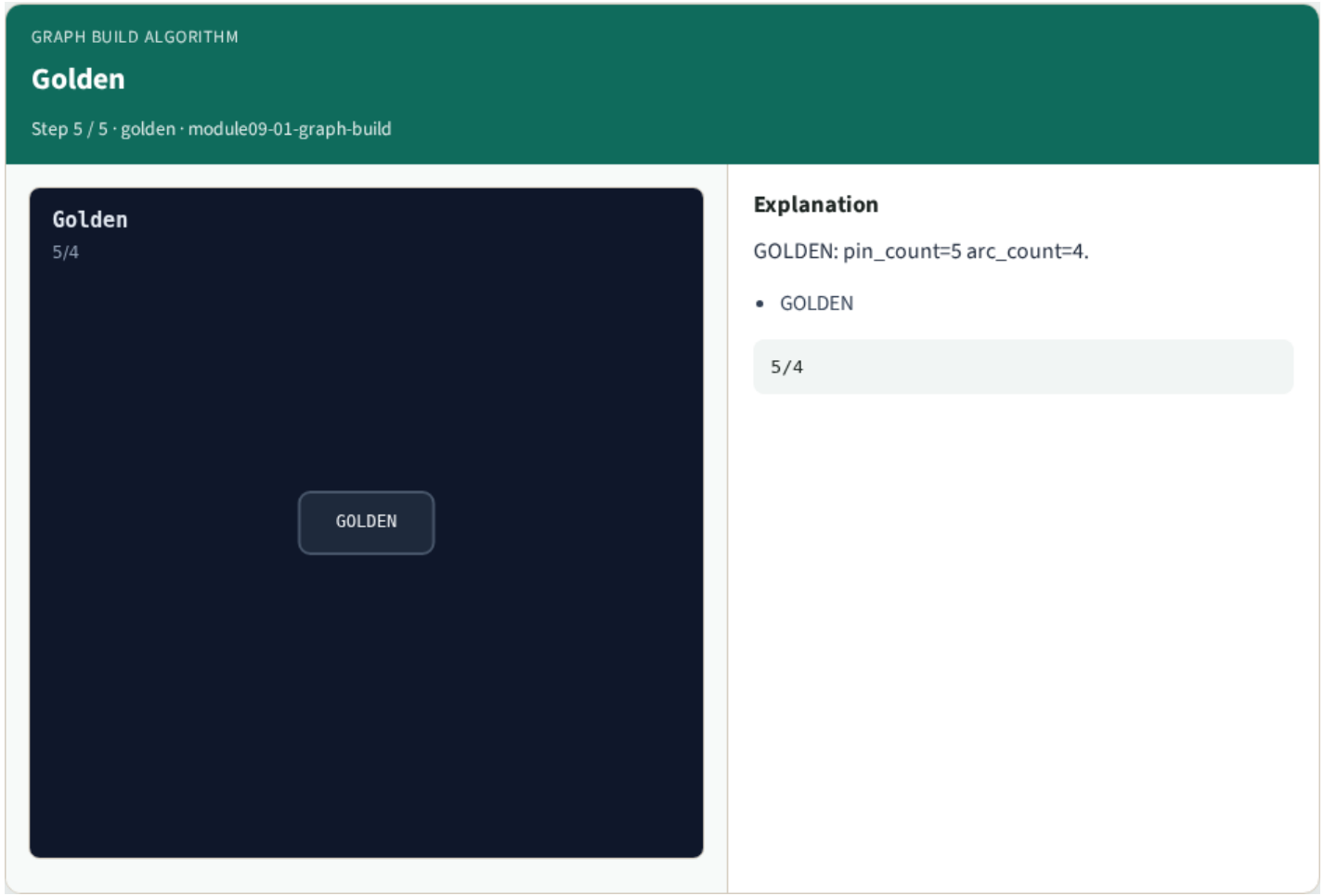
Cells

Graph


GRAPH BUILD ALGORITHM
Graph
Step 4 / 5 · graph · module09-01-graph-build
Graph
ok
graph
Explanation
The timing graph is pins plus arcs.
graph
ok

Graph

Golden



Golden

Browser lab track

- Open **graph-build**
- Run Graph / Levelize / Propagate / Slack / Bind, then Check

Implement track

- From common/, run algo_solvers.py and partc_bind.py

Pitfalls

- Changing tiny_algo_sta without updating Part B liberty delays
- Ignoring the bind side

Try these

- From the common folder, run the Part C self-tests

Try these — try these

```
cd courses/learn_opensta/common
python3 algo_solvers.py
python3 partc_bind.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
5 pins, 4 arcs; cells u1/u2/u3.
ok if goldens match and bind
```

Next

- Complete the quiz for this part
- Continue Part C, then wrap , internals literacy is next